RAFIUL RABBI ZAHID

IoT Enthusiast | Artificial Intelligence Researcher

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LinkedIn | GitHub | Portfolio | ORCID

Education

Bachelor of Science in Internet of Things and Robotics Engineering April 2022 – September 2026

University of Frontier Technology, Bangladesh | CGPA: 3.52/ 4.00

Higher Secondary Certificate 2018 – 2020

Government Edward College, Pabna | GPA: 5.00 / 5.00

Secondary School Certificate 2018

Pabna Zilla School | GPA: 5.00 / 5.00

Experience

Research Assistant Dhaka, Bangladesh | September 2026 – Present

Department of Electrical and Computer Engineering – North South University

- Conduct research on AI, machine learning, and IoT-driven intelligent systems, focusing on system development, experimentation, data analysis, and research publication.

Research Assistant Remote | May 2026 – August 2026

Elite Research Lab LLC

- Conduct applied research across AI, computer vision, NLP, and edge AI, producing publication-ready experimental results for peer-reviewed venues.

Technical Skills

Programming Languages: Python, C, C++

Machine Learning & Deep Learning: PyTorch, Transfer Learning, CNN Architecture Design.

NLP & LLM: Transfer Learning, Retrieval-Augmented Generation (RAG), LLM Integration, Vector Search

Data & Analytics: NumPy, Pandas, Matplotlib, Data Preprocessing, Exploratory Data Analysis

Databases: MySQL, SQLite, FAISS, Chroma (Vector Databases)

Embedded & IoT Systems: Arduino, ESP32, Sensor Integration, Actuator Control

AI & Automation Tools: ChatGPT, Claude, n8n Workflow Automation

Platforms & Tools: Linux, Windows, LaTeX, Microsoft Office

Research Interests

- **Core Areas:** Artificial Intelligence, Internet of Things, Explainable AI, NLP, Computer Vision
- **Research Focus:** AI for Healthcare, Edge Intelligence, and Resource-Efficient Learning
- **Applications:** Smart Agriculture, Healthcare AI, Robotics and Autonomous Systems

B.Sc Thesis Title

A Lightweight Diffusion–Transformer Framework with Knowledge Distillation for Uncertainty-Aware Diabetic Retinopathy Detection

Projects

IoT and ML-Enabled Smart Vertical Farming System July 2025

Capstone Project | ESP32, Machine Learning, Sensors

Developed an ESP32-based monitoring system with ML-based crop recommendations using environmental data.

IoT-Enabled Autonomous Monitoring Rover April 2024

Robotics & IoT Project | ESP32, ROS, LiDAR, IMU, Encoders

Developed an ESP32 rover with LiDAR, IMU, and encoders for sensing, odometry, and ROS-based navigation.

Multilingual Toxic Content Detection System February 2026
Course Project | Python, PyTorch, mBERT, XLM-RoBERTa, Flask

Developed a transformer-based multi-label classifier for Bangla and Banglish toxicity detection.

RAG Healthcare Assistant January 2025
NLP & Generative AI Project | RAG, Hugging Face, FAISS

Developed a RAG-based healthcare assistant with retrieval and reranking for evidence-based question answering.

Intelligent Email Response Automation System May 2026
AI & Workflow Automation Project | n8n, Gmail API, OpenAI

Automated Gmail classification and AI-assisted response generation using n8n and OpenAI.

Weather Irrigate Hub — Smart Irrigation System July 2023
IoT Project | Arduino UNO, DHT11, Soil Moisture Sensor

Developed a sensor-driven irrigation system with automated actuator control using Arduino.

Publications

Journal Articles

- **Rafiul R. Zahid**, Rakibul H. Rakib, Iftekhar Rahman, Md. A. Amin, Tanjum R. Akanto, Fahmida A. Antara, Sadia Enam, "IoT for Industrial Automation: A Review on Industry 4.0 Applications." *International Journal of Information Engineering and Electronic Business (IJIEEB)*, MECS Press, 2026. DOI: 10.5815/ijieeb.2026.02.11

Conference Papers

- Rakibul Hasan Rakib, **Rafiul Rabbi Zahid**, Suman Saha, "VertiCulture: An Integrated IoT and Machine Learning Enabled Smart System for Optimizing Vertical Farming Environments." In *Proceedings of the 2026 IEEE 2nd International Conference on Quantum Photonics, Artificial Intelligence & Networking (QPAIN)*, IEEE, 2026.
- Abidul Islam Alif, Saadman Jawad Naveed, **Rafiul Rabbi Zahid**, Md. Rafiqul Islam, Md. Tamzid Hossen Omi, Afsana Mim, "WatchRover: An IoT-Enabled Autonomous Bot for Disaster Response and Rough Terrain Navigation." In *Proceedings of the 2026 IEEE 2nd International Conference on Quantum Photonics, Artificial Intelligence & Networking (QPAIN)*, IEEE, 2026.
- **Rafiul Rabbi Zahid**, Abidul Islam Alif, Shahbaj Ahmed Sakib, Tajbir Ahmed, Abdul Hasib, Mostafiz Ahammed, "Environmental Drivers of Lumpy Skin Disease: An Interpretable Ensemble Machine Learning Framework for Climatic Risk Assessment." *International Conference on Electrical, Computer and Communication Technologies (ECCT)*, 2026. **(Accepted)**
- **Rafiul Rabbi Zahid**, Rakibul Hasan Rakib, Abidul Islam Alif, Iftekhar Rahman, Shahbaj Ahmed Sakib, Abdul Hasib, "Explainable Machine Learning Framework for Symptom-Based Disease Prediction with Robustness Analysis." *International Conference on Electrical, Computer and Communication Technologies (ECCT)*, 2026. **(Accepted)**
- Abidul Islam Alif, **Rafiul Rabbi Zahid**, Saadman Jawad Naveed, Muzahidul Islam Saki, Md. Tanjim Khan, Abdul Hasib, "Deep Learning-Based Hourly Traffic Volume Forecasting Using Stacked LSTM Networks: A Multi-Junction Evaluation on Heterogeneous Urban Road Sensors." *International Conference on Electrical, Computer and Communication Technologies (ECCT)*, 2026. **(Accepted)**
- **Rafiul Rabbi Zahid**, Abidul Islam Alif, Shahbaj Ahmed Sakib, Maria Akter Mim, Md. Abu Horaira Ansary Tanvir, Abdul Hasib, "An Explainable and Uncertainty-Aware Machine Learning Framework for Automated Smoking Risk Assessment Using Clinical Health Indicators." *International Conference on Power, Electronics, Communications, Computing, and Intelligent Infrastructure (PECCII 2026)*, Pabna University of Science and Technology (PUST), 2026.
- **Rafiul Rabbi Zahid**, Rakibul Hasan Rakib, Iftekhar Rahman, Shahbaj Ahmed Sakib, Suman Saha, "An Explainable and Uncertainty-Aware Toxicity Detection in Low-Resource Multilingual Text Using Transformers." *International Conference on Computing Advancements (ICCA 2026)*, AIUB, 2026. **(Accepted)**
- Avik Halder, **Rafiul Rabbi Zahid**, Rakibul Hasan Rakib, Saurav Chandra Das, Sadia Enam, Md. Toukir Ahmed, Tanjum Rahi Akanto, Fahmida Ahmed Antara, "AgroLux: A Low-Cost Sustainable Solar-Driven Smart Irrigation Framework with Adaptive Sensing." In *Proceedings of the IEEE 3rd International Conference on Computing, Applications and Systems (COMPAS)*, IEEE, 2026.

Journal Manuscripts Under Review

- **Rafiul Rabbi Zahid**, Shahbaj Ahmed Sakib, Abidul Islam Alif, Md. Toukir Ahmed, Fahmida Ahmed Antara, "Edge AI for the Internet of Things: A Comprehensive Systematic Review of Architectures, Algorithms, and Applications." 2026. **(Under Review)**
- **Rafiul Rabbi Zahid**, Avik Halder, Mostafiz Ahammed, Farzana Akter, Adila Nuzhat, Abu Sayed, Md. Mostafizur Rahaman, "A Systematic Review of IoT Systems Testing: A Taxonomy and Maturity Framework for Cross-Layer Validation and Automation." 2026. **(Under Review)**
- Ahmed Faizul Haque Dhrubo, Md. Mofazzal Hossen, **Rafiul Rabbi Zahid**, Asmaul Islam Tisha, Most Aysha Siddika Sumona, Mohsin Sajjad, Acramul Haque Kabir, Mohammad Abdul Qayum, Mohammad Ashrafuzzaman Khan, "Automated Vision-Based Corrosion Inspection for Steel Infrastructure Using Lightweight Adaptive Deep Learning." 2026. **(Under Review)**
- Rakibul Hasan Rakib, **Rafiul Rabbi Zahid**, Suman Saha, "A Lightweight Diffusion–Transformer Framework with Knowledge Distillation for Uncertainty-Aware Diabetic Retinopathy Detection." 2026. **(Under Review)**

Achievements & Awards

- Dean's List Award, Faculty of Cyber Physical Systems Engineering, UFTB – Outstanding Academic Performance, Level 4.
- Dean's List Award, Faculty of Cyber Physical Systems Engineering, UFTB – Outstanding Academic Performance, Level 3.
- 1st Runner-Up, IoTrix – Televerse 1.0 National-Level Hackathon, organized by Dept. of ETE, CUET.
- 2nd Runner-Up, Espectrum 1.0 Project Showcasing Event, organized by RUET IoT Club .

Leadership & Professional Involvement

- Executive Member, Basis Student Forum – UFTB Chapter (March 2024 – June 2025).
- Executive Member, UFTB Debating Society (May 2023 – August 2024).
- Student Activity Member, UFTB Cloud Club (November 2023 – August 2024).

Certifications

- 21st Century Employability Skilling Program — Wadhwani Foundation (February 2024)

References

Suman Saha
Assistant Professor
Department of Internet of Things and Robotics Engineering
University of Frontier Technology, Bangladesh
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Relationship: Thesis Supervisor

Fahmida Ahmed Antara
Assistant Professor
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Phone: +8801924470784
Relationship: Course Instructor